

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I M. Sandhya (192312421) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.
Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

ANSWERS

1 SOLUTION

Types of Intelligent Agents in Smart Home Automation

1. Simple Reflex Agent

- Works on **condition–action rules** (if–then).
- Uses only **current input**, no memory.

In the system:

- If motion detected → turn ON lights

- If temperature rises → turn ON AC
- If door opens at night → trigger alarm

Analysis:

- Fast and suitable for **real-time response**
- Cannot learn or adapt to user behavior
- Not suitable alone for a smart home

2. Model-Based Agent

- Maintains an **internal state (memory)** of the environment.

In the system:

- Remembers which rooms are occupied
- Tracks time (day/night) and previous sensor data
- Avoids wrong actions (e.g., turning off lights when someone is inside)

Analysis:

- Handles **partial information** better
- More reliable than reflex agents
- Still limited because it doesn't plan or optimize

3. Goal-Based Agent

- Takes actions to **achieve specific goals**.

In the system:

- Goal: Maintain comfortable temperature
- Goal: Ensure home security
- Decides actions like adjusting AC gradually or activating security modes

Analysis:

- More flexible and intelligent
- Can evaluate different actions
- Does not consider **which solution is best** among alternatives

4. Utility-Based Agent

- Uses a **utility function** to measure how good a state is.
- Chooses actions that **maximize overall benefit**.

In the system:

- Balances:
 - Comfort (lighting, temperature)
 - Energy efficiency
 - Security

- Example: Reduce AC slightly to save energy without affecting comfort

Analysis:

- Best for **decision-making with trade-offs**
- Provides optimal and adaptive behavior
- More complex to design

Best Approach (Justification)

A **hybrid approach** is most effective:

- **Simple Reflex** → handles immediate real-time actions
- **Model-Based** → provides context awareness
- **Goal-Based** → ensures objectives (comfort, security)
- **Utility-Based** → optimizes decisions (energy vs comfort vs safety)

Conclusion

A **combination of model-based, goal-based, and utility-based agents with reflex rules** is ideal.

This hybrid system ensures:

- **Comfort** through personalized responses
- **Energy efficiency** by reducing unnecessary usage
- **Security** through intelligent monitoring

Hence, a **utility-based hybrid intelligent agent** is the most effective solution for a smart home automation system.

2 SOLUTION

Robot in Unknown Maze – Uninformed Search Strategies

A robot in an unknown maze must explore paths step by step without prior knowledge. **Uninformed search algorithms** like **Uniform Cost Search (UCS)** and **Iterative Deepening Search (IDS)** help the robot systematically find the exit.

1. Uniform Cost Search (UCS)

Concept:

- Expands the node with the **lowest path cost**.
- Uses a priority queue.

In the maze:

- Robot explores paths based on **minimum travel cost** (e.g., distance, time).
- Always guarantees the **least-cost path to the exit**.

Advantages:

- Complete (will find a solution if it exists)
- ☒ Optimal (finds lowest-cost path)

Disadvantages:

- High **time complexity**: $O(bd)O(b^{\{d\}})O(bd)$ (can explore many nodes)
- High **space complexity**: stores all generated nodes $\rightarrow$ memory-intensive
- Slow when path cost differences are small

2. Iterative Deepening Search (IDS)

Concept:

- Combines **Depth-First Search + Breadth-First Search**
- Repeats DFS with increasing depth limits

In the maze:

- Robot explores paths up to depth 1, then 2, then 3, etc.
- Eventually finds the exit at the **shallowest level**

Advantages:

- Complete
- Optimal (for equal step costs)
- Low **space complexity**: $O(bd)O(bd)O(bd) \rightarrow$ very memory efficient

Disadvantages:

- Repeats exploration $\rightarrow$ higher **time overhead**
- Not suitable when costs vary (no cost awareness)

Comparison (Time & Space)

Algorithm	Time Complexity	Space Complexity	Optimal
UCS	High $O(bd)O(b^d)O(bd)$	Very High	Yes
IDS	Moderate - High	Low $O(bd)O(bd)O(bd)$	Yes (unit cost)

Relation to Intelligent Agents

1. Simple Reflex Agent

- Acts only on current perception
- Cannot handle maze (no memory, no planning)

2. Model-Based Agent

- Maintains internal map of explored paths
- Useful for avoiding revisiting same locations

3. Goal-Based Agent

- Has goal: **reach exit**
- Uses search algorithms (UCS/IDS) to plan path

4. Utility-Based Agent

- Evaluates best path based on:
 - Cost (distance/time)
 - Risk (dead ends)
- Chooses most efficient solution

Effect of Agent Design on Decision-Making

- **Without memory (reflex)** → random movement, inefficient
- **With model (memory)** → avoids loops, better exploration
- **Goal-based** → purposeful search using algorithms
- **Utility-based** → selects optimal path considering trade-offs

Better agent design = **faster, smarter decisions**

Best Combination (Justification)

Recommended: Goal-Based + Model-Based Agent with UCS (Hybrid)

Why?

- **Goal-Based** → ensures robot focuses on reaching exit
- **Model-Based** → remembers explored paths (avoids repetition)
- **UCS** → guarantees **optimal (least-cost) path**

Alternative:

- Use **IDS** when:
 - Memory is limited
 - All steps have equal cost

Final Conclusion

- **UCS** is best for **optimal path finding** but uses more memory
- **IDS** is best for **memory efficiency** but less cost-aware
- A **Goal-based + Model-based agent using UCS** is most effective because it:
 - Plans intelligently
 - Avoids repeated exploration
 - Finds the best path efficiently

Thus, a **hybrid intelligent agent with UCS** provides the best balance of **accuracy, efficiency, and decision-making** in an unknown maze.